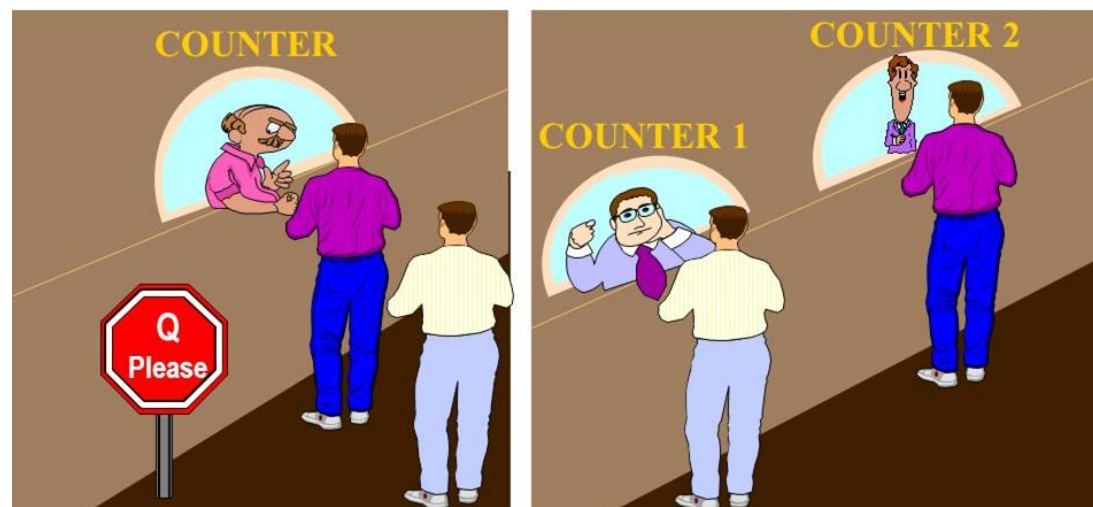


Basics of Parallel & Distributed Computing

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OUTLINE

- Why Parallel and Distributed Computing Curriculum?
- What is Parallel Computing – videos
- Unplugged activity 1: Sorting Pennies
 - Activity phases: sequential sorting, parallel sorting
 - Group/class discussion
 - Survey
- Plugged activity: Sorting Pennies using code
 - Sequential sorting
 - Parallel sorting
 - Survey

Why a Parallel and Distributed Computing Curriculum?

An initiative mandated by the National Science Foundation (NSF) and the ACM/IEEE Computer Society

https://grid.cs.gsu.edu/~tcpp/curriculum/?q=curr_why_pdc.html

What is Parallel Computing?

- Parallel Computing: [What is parallel computing?](#)
- [Sequential vs Parallel vs Distributed Computing Explained Simply](#)

More videos for later:

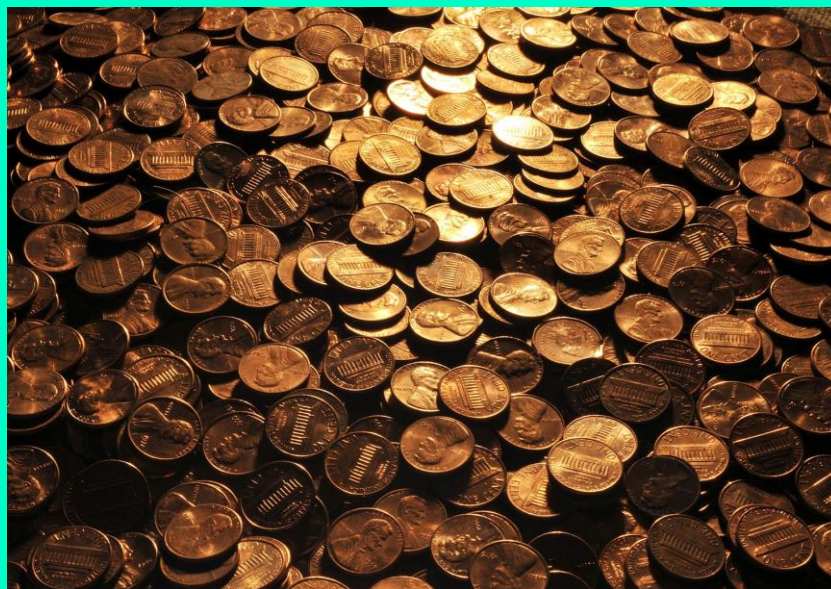
<https://www.youtube.com/watch?v=RJNx90ODawk>

<https://www.youtube.com/watch?v=GXrI3oJ3Eak>

- Description

<https://www.britannica.com/science/computer-science/Parallel-and-distributed-computing>

UNPLUGGED ACTIVITY 1: A PILE OF PENNIES



UNPLUGGED ACTIVITY 2: DESCRIPTION

Goal: find oldest penny.

We are going to explore several solutions to achieving the goal.

UNPLUGGED ACTIVITY 2 DESCRIPTION

I have 50 coins and the problem is to find the oldest one in the quickest time. So we are going to do it first in a sequential mode and then in various parallel variations.

Students break into groups of 5 or 6.

Phase 1 (sequential): each group (student 1 sorting and student 2 timing, other students observing)

Student 1 task: find the oldest penny

Student 2 task: time the activity of student 1 and writes it down. Other students' task: observe Student 1 and 2. When they are done **Shuffle coins and put back in bag**

UNPLUGGED ACTIVITY 2 DESCRIPTION CONT'D....

Phase 2 (Parallelize it): each group (students 1 and 2 are given equal number of pennies and are asked to find the oldest penny in their pile, student 3 records the times for student 1 & 2. Parallel Time is calculated as the time of the slowest student among 1 and 2.

At a set time, sorting begins for all groups.

Student 3 records the time it takes each student to find the oldest penny in their pile. Parallel Time is calculated as the time of the slowest student among 1 and 2.

When done shuffle coins and put back in their respective bags. Please do not mix coins. Each bag should have 25 pennies.

UNPLUGGED ACTIVITY 2 DESCRIPTION CONT'D....

Phase 3 (Parallelize it-2):

Parallelize it (2): Teacher gives 2 pennies to students 1, 2, 3, 4 and the remaining pennies to student 5.

At a set time, sorting begins for all groups.

Student 6 records time for each of the worker students. Parallel time is calculated as the time of the slowest student.

What do we learn? **This is poor load balancing!** *Shuffle and put coins back in the bag*

Group Discussion:
what did we learn?

PLUGGED (CODE) ACTIVITY 1: SORTING AN ARRAY OF NUMBERS

2010	1965	2013	1943	1910	2009	1942	1903	2018
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Review Code

- Review Code for sequential sorting – PennySortingSequential.java
- Review Code for parallel sorting - PennySortingParallel.java
 - parallelSort()
 - Sorting with Stream
 - Fork/Join

STUDENT SURVEY

- Student Engagement Survey – link provided

Thank you for your
participation!

Hope you enjoyed the
parallelism day